

ONLINE BUSINESS AND TECHNOLOGY

Every business is now, at least partly, a technology business. Today we look at how companies turn raw data into decisions, keep that data safe, and adapt to tools — social, mobile, and AI — that keep changing the rules.

WHAT YOU'RE GETTING FOR YOUR 50 MINUTES

- 10.1** Describe how a business transforms data into useful business intelligence.
- 10.2** Understand the security and privacy challenges of advancing technology.
- 10.3** Explain how social networking and mobile technologies have impacted business.
- 10.4** Understand how new advances in technology impact business.

Gold-outlined boxes are in-class activities — that's your cue to actually participate, not just read.

THE DEFINITION WE BUILD ON ALL CLASS

Business intelligence. The organized, analyzed information a company uses to make better decisions — built from raw data.

Chapter 9 was about people. Chapter 10 is about the tools those people now rely on to do the job.

OBJECTIVE 10.1

From raw data to smart decisions

A spreadsheet full of numbers isn't worth much until someone turns it into a decision.

Two words people mix up

Data. Raw, unorganized facts — on their own, not very useful.

YOUR OWN EXAMPLE

Information. Data that has been organized in a way that makes it useful for decision-making.

YOUR OWN EXAMPLE

Business intelligence tools

Tap each to see what it does.

01 Databases

Organized collections of data that can be searched and updated.

EX.

02 Data warehouses

Large central repositories that combine data from many sources across a company.

EX.

03 Data mining

Searching large datasets for patterns and relationships that aren't obvious at first glance.

EX.

04 Big Data analytics

Analyzing extremely large, fast-moving datasets to find insights in near real time.

EX.

05 CRM systems

Customer relationship management software that tracks every interaction a company has with a customer.

EX.

06 DSS & EIS systems

Decision support and executive information systems that turn data into ready-to-use insight for managers.

EX.

REAL-WORLD EXAMPLE

Netflix's recommendation engine is data mining in action — it studies what millions of people watch to predict what you'll want to watch next.

NOTES · 10.1 FROM RAW DATA TO SMART DECISIONS

OBJECTIVE 10.2

Keeping systems — and people — safe

The more a business runs on data, the more it has to protect, and the more it can be tempted to watch.

Security threats

- Hackers
- Phishing
- Spyware
- Adware
- Ransomware

Security measures

- Password protection
- Two-factor authentication (2FA)
- Biometrics
- Passkeys

Privacy concerns

Employee monitoring. Tracking employee activity, communications, or location — legal in many contexts, but a real tension with trust and privacy.

YOUR OWN EXAMPLE

Customer data protection. A company's responsibility to secure and responsibly use the personal data customers hand over.

YOUR OWN EXAMPLE

IN-CLASS DISCUSSION

Where should the line be between monitoring employees for security and respecting their privacy?

NOTES · 10.2 KEEPING SYSTEMS — AND PEOPLE — SAFE

OBJECTIVE 10.3

Business goes social and mobile

Social platforms and phones didn't just change how people talk — they changed how businesses market, hire, and sell.

What businesses use social media for

01 Marketing

Reaching customers directly, often for a fraction of traditional ad costs.

EX. _____

02 Customer service

Answering questions and resolving complaints in public view.

EX. _____

03 Recruiting

Finding and vetting job candidates.

EX. _____

04 Brand awareness

Building visibility and reputation over time.

EX. _____

Two ways the crowd gets involved

Crowdsourcing. Getting ideas, content, or work from a large group of people, often online.

YOUR OWN EXAMPLE

Crowdfunding. Raising small amounts of money from a large number of people, usually online.

YOUR OWN EXAMPLE

Social media monitoring. Tracking mentions and conversations about a brand across social platforms.

YOUR OWN EXAMPLE

Mobile technology

- Mobile apps
- Mobile commerce
- Cloud computing

DISCUSSION QUESTION

Name a brand that uses social media particularly well. What do they do differently from everyone else?

NOTES · 10.3 BUSINESS GOES SOCIAL AND MOBILE

OBJECTIVE 10.4

AI and what comes next

Artificial intelligence has moved from science fiction to a tool most businesses already use in some form.

Artificial Intelligence (AI). Technology that performs tasks normally requiring human intelligence.

YOUR OWN EXAMPLE

Where businesses already use it

01 Automation

Handling repetitive tasks without a human doing them manually.

EX. _____

02 Customer support

Chatbots and virtual assistants that handle common questions instantly.

EX. _____

03 Data analysis

Finding patterns in data far faster than a person could.

EX. _____

04 Content creation

Drafting text, images, or code to speed up creative and technical work.

EX. _____

Two terms people often confuse

Machine learning. A type of AI that learns patterns from data and improves its performance over time, without being explicitly reprogrammed.

YOUR OWN EXAMPLE

Generative AI. AI that creates new text, images, code, or other content, rather than just analyzing existing data.

YOUR OWN EXAMPLE

What's coming next

- Quantum computing
- Increasingly capable AI-powered business tools

DISCUSSION QUESTION

Where have you already used generative AI this week? Where do you think businesses should draw an ethical line on how they use it?

NOTES · 10.4 AI AND WHAT COMES NEXT

CHAPTER 10 · WRAP-UP

Closing the terminal

Define each term in your own words. If you can do all of them without looking, you're ready for the quiz.

DEFINE THESE IN YOUR OWN WORDS

- | | | |
|---------------------------|-----------------------------|-------------------|
| • Data | • Information | • Data mining |
| • Big Data | • CRM | • Phishing |
| • Ransomware | • Two-factor authentication | • Crowdsourcing |
| • Crowdfunding | • Mobile commerce | • Cloud computing |
| • Artificial Intelligence | • Machine learning | • Generative AI |

ACTIVITY MINUTE · BEFORE YOU LEAVE

Which key term from this chapter matters most, and why?

Your one biggest “aha” moment from today.

What technology do you think will change business the most in the next five years?

TOTAL — 50 MIN · PAID IN FULL

Chapter 9 was about the people running the business. Chapter 10 was about the tools they now rely on. Next, we follow a product from raw materials to a customer's hands.